



Top 54 CAT Logarithms, Surds and Indices Questions With Video Solutions

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Questions

Instructions

For the following questions answer them individually

Question 1

If x and y are positive real numbers such that $\log_x(x^2 + 12) = 4$ and $3 \log_y x = 1$, then $x + y$ equals

- A 20
- B 68
- C 10
- D 11

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If x and y are positive real numbers such that $\log_x(x^2 + 12) = 4$ and $3 \log_y x = 1$, then $x + y$ equals

A) 20
B) 68
C) 10
D) 11

$x^2 + 12 = x^4$
 $x^4 - x^2 = 12$
 $x^2(x^2 - 1) = 12$
 $x^2(x - 1)(x + 1) = 12$
 $x^2 = 4$
 $x = 2$
 $3 \log_y 2 = 1$
 $\log_y 2 = \frac{1}{3}$
 $2 = y^{\frac{1}{3}}$
 $2^3 = y$
 $y = 8$
 $x + y = 2 + 8 = 10$



VIDEO SOLUTION

Question 2

If $(a + b\sqrt{3})^2 = 52 + 30\sqrt{3}$, where a and b are natural numbers, then $a + b$ equals

- A 7
- B 8
- C 9
- D 10

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If $(a + b\sqrt{3})^2 = 52 + 30\sqrt{3}$, where a and b are natural numbers, then $a + b$ equals

A) 7
B) 8
C) 9
D) 10

$(a + b\sqrt{3})^2 = 52 + 30\sqrt{3}$
 $a^2 + 3b^2 + 2ab\sqrt{3} = 52 + 30\sqrt{3}$
 $a^2 + 3b^2 = 52$
 $2ab = 30$
 $ab = 15$
 $a = 3, b = 5$
 $a + b = 8$



VIDEO SOLUTION

Question 3

Let $u = (\log_2 x)^2 - 6\log_2 x + 12$ where x is a real number. Then the equation $x^u = 256$, has

- A no solution for x
- B exactly one solution for x
- C exactly two distinct solutions for x
- D exactly three distinct solutions for x

Handwritten solution for Question 3:

Let $u = (\log_2 x)^2 - 6\log_2 x + 12$ where x is a real number. Then the equation $x^u = 256$, has

A) no solution for x
B) exactly one solution for x
C) exactly two distinct solutions for x
D) exactly three distinct solutions for x

$\log_2 x = A$

$u = A^2 - 6A + 12$

$x^u = 256$

$u = \log_x 256$

$8 \log_x 2$

$\frac{8}{A}$

VIDEO SOLUTION

CAT Mock Test

Mocks designed exactly like the actual CAT

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Question 4

If N and x are positive integers such that $N^N = 2^{160}$ and $N^2 + 2^N$ is an integral multiple of 2^x , then the largest possible x is

Handwritten solution for Question 4:

If N and x are positive integers such that $N^N = 2^{160}$ and $N^2 + 2^N$ is an integral multiple of 2^x , then the largest possible x is

$N^N = 2^{160}$

$N = 2^{160/N}$

$N = 2^{40}$

$N = 16$

$N^2 + 2^N = 16^2 + 2^{16}$

$256 + 65536$

65792

$2^8 \times 1031$

2^8

VIDEO SOLUTION

Question 5

If $\log(2^a \times 3^b \times 5^c)$ is the arithmetic mean of $\log(2^2 \times 3^3 \times 5)$, $\log(2^6 \times 3 \times 5^7)$, and $\log(2 \times 3^2 \times 5^4)$, then a equals

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If $\log(2^a \times 3^b \times 5^c)$ is the arithmetic mean of $\log(2^2 \times 3^3 \times 5)$, $\log(2^6 \times 3 \times 5^7)$, and $\log(2 \times 3^2 \times 5^4)$, then a equals

$$AM = \frac{\log(2^2 \times 3^3 \times 5) + \log(2^6 \times 3 \times 5^7) + \log(2 \times 3^2 \times 5^4)}{3}$$

$$= \frac{1}{3} \times \log(2^{2+6+1} \times 3^{3+1+2} \times 5^{1+7+4})$$

$\log A + \log B = \log(A \times B)$



VIDEO SOLUTION

Question 6

If m and n are integers such that $(\sqrt{2})^{19} 3^4 4^2 9^m 8^n = 3^n 16^m (\sqrt[4]{64})$ then m is

- A -20
- B -24
- C -12
- D -16

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If m and n are integers such that $(\sqrt{2})^{19} 3^4 4^2 9^m 8^n = 3^n 16^m (\sqrt[4]{64})$ then m is.

$$LHS = \frac{2^{19}}{2} \times \frac{3^4}{3} \times \frac{2^4}{2} \times \frac{3^{2m}}{3} \times \frac{2^{3n}}{2} = 2^{18} \times 3^{4-1} \times 2^{3-1} \times 3^{2m-1} \times 2^{3n-1}$$

$$RHS = 3^n \times 2^{4m} \times 2^3$$


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Question 7

If $\sqrt{5x+9} + \sqrt{5x-9} = 3(2 + \sqrt{2})$, then $\sqrt{10x+9}$ is equal to

- A $3\sqrt{31}$
- B $4\sqrt{5}$
- C $3\sqrt{7}$

D $2\sqrt{7}$

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If $\sqrt{5x+9} + \sqrt{5x-9} = 3(2+\sqrt{2})$, then $\sqrt{10x+9}$ is equal to

A) $3\sqrt{31}$ B) $4\sqrt{5}$
C) $3\sqrt{7}$ D) $2\sqrt{7}$

$\sqrt{5x+9} = k - \sqrt{5x-9}$
 $5x+9 = k^2 - 2\sqrt{5x-9} + 5x-9$
 $2\sqrt{5x-9} = k^2 - 18$

VIDEO SOLUTION

Question 8

For some positive real number x , if $\log_{\sqrt{3}}(x) + \frac{\log_x(25)}{\log_x(0.008)} = \frac{16}{3}$, then the value of $\log_3(3x^2)$ is

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For some positive real number x , if $\log_{\sqrt{3}}(x) + \frac{\log_x(25)}{\log_x(0.008)} = \frac{16}{3}$, then the value of $\log_3(3x^2)$ is

Answer: _____

$\log_x 25 = \frac{\log 25}{\log x}$
 $\log_x 0.008 = \frac{\log 0.008}{\log x}$
 $\frac{\log 25}{\log x} \div \frac{\log 0.008}{\log x} = \frac{\log 25}{\log 0.008} = \frac{2.5}{-3} = -\frac{5}{6}$

VIDEO SOLUTION

Question 9

$\frac{2 \times 4 \times 8 \times 16}{(\log_2 4)^2 (\log_4 8)^3 (\log_8 16)^4}$ equals

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$\frac{2 \times 4 \times 8 \times 16}{(\log_2 4)^2 (\log_4 8)^3 (\log_8 16)^4}$ equals

Denominator = $2^2 \times 2^3 \times 2^4$

$\log_2 8 = \frac{3}{2}$
 $\log_4 8 = \frac{3}{2} \times \log_2 2 = \frac{3}{2}$
 $\log_x x^m = m \times \log_x x$
 $\log_y x = \frac{1}{m} \times \log_y x$

VIDEO SOLUTION



Question 10

The sum of all distinct real values of x that satisfy the equation $10^x + \frac{4}{10^x} = \frac{81}{2}$, is

- A $2 \log_{10} 2$
- B $4 \log_{10} 2$
- C $\log_{10} 2$
- D $3 \log_{10} 2$

The sum of all distinct real values of x that satisfy the equation $10^x + \frac{4}{10^x} = \frac{81}{2}$, is

A) $2 \log_{10} 2$
B) $4 \log_{10} 2$
C) $\log_{10} 2$
D) $3 \log_{10} 2$

$10^x = A$

$A + \frac{4}{A} = \frac{81}{2}$

$A^2 - \frac{81}{2}A + 4 = 0$

$A_1 \times A_2 = 4$
 $A_1 + A_2 =$



VIDEO SOLUTION

Question 11

If $3^a = 4$, $4^b = 5$, $5^c = 6$, $6^d = 7$, $7^e = 8$ and $8^f = 9$, then the value of the product $abcdef$ is

If $3^a = 4$, $4^b = 5$, $5^c = 6$, $6^d = 7$, $7^e = 8$ and $8^f = 9$, then the value of the product $abcdef$ is

Answer: _____

$a = \log_3 4$
 $b = \log_4 5$
 $c =$
 $d =$

$e = \log_7 8$
 $f = \log_8 9$



VIDEO SOLUTION

Question 12

If $5 - \log_{10} \sqrt{1+x} + 4 \log_{10} \sqrt{1-x} = \log_{10} \frac{1}{\sqrt{1-x^2}}$, then $100x$ equals



VIDEO SOLUTION



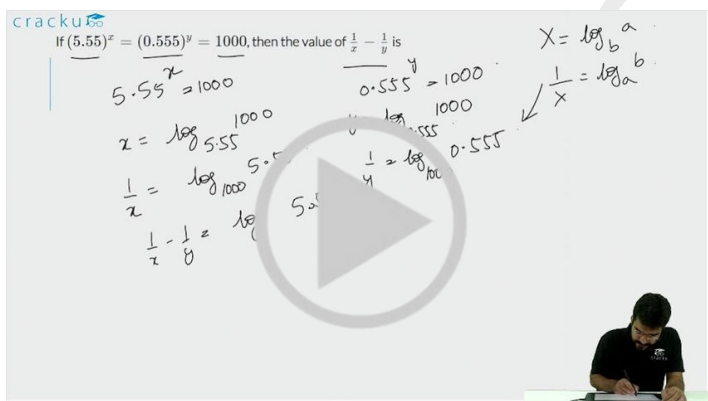
CAT Syllabus PDF



Question 13

If $(5.55)^x = (0.555)^y = 1000$, then the value of $\frac{1}{x} - \frac{1}{y}$ is

- A $\frac{1}{3}$
- B 3
- C 1
- D $\frac{2}{3}$



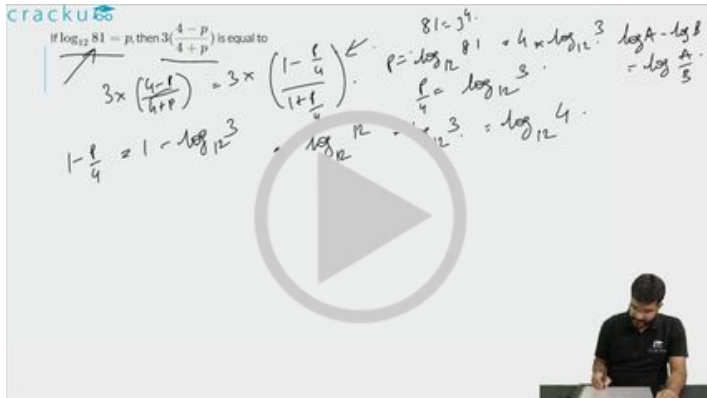
VIDEO SOLUTION

Question 14

If $f(x) = \log \frac{(1+x)}{(1-x)}$, then $f(x) + f(y)$ is

- A $f(x + y)$
- B $f\left(\frac{x+y}{1+xy}\right)$
- C $(x + y)f\left(\frac{1}{1+xy}\right)$

- B $\log_6 16$
- C $\log_2 8$
- D $\log_6 8$

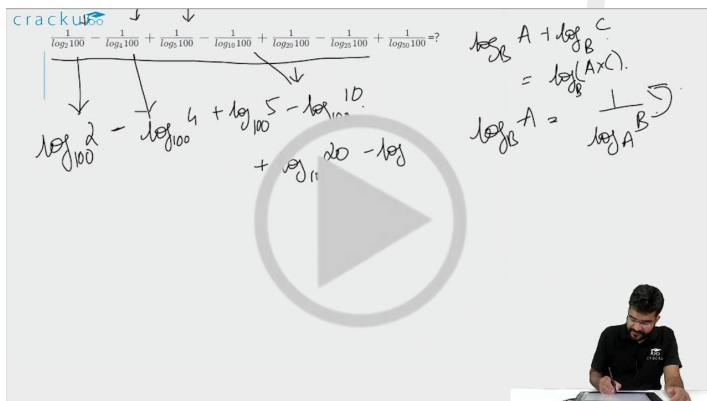


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Question 17

$$\frac{1}{\log_2 100} - \frac{1}{\log_4 100} + \frac{1}{\log_5 100} - \frac{1}{\log_{10} 100} + \frac{1}{\log_{20} 100} - \frac{1}{\log_{25} 100} + \frac{1}{\log_{50} 100} = ?$$

- A $\frac{1}{2}$
- B 10
- C 0
- D -4



VIDEO SOLUTION

Question 18

If x is a positive quantity such that $2^x = 3^{\log_5 2}$, then x is equal to

- A $\log_5 8$
- B $1 + \log_3(\frac{5}{3})$
- C $\log_5 9$

D $1 + \log_5\left(\frac{3}{5}\right)$

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If x is a positive quantity such that $2^x = 3^{\log_5 2}$, then x is equal to

A) $\log_5 8$
B) $1 + \log_5\left(\frac{3}{5}\right)$
C) $\log_5 9$
D) $1 + \log_5\left(\frac{3}{2}\right)$

Handwritten solution: $2^x = 3^{\log_5 2}$
Taking log on both sides:
 $x \log 2 = \log_5 2 \cdot \log 3$
 $x = \frac{\log 3}{\log 2} \cdot \log_5 2$
 $x = \log_5 3$

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VIDEO SOLUTION

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Accurate analysis of your profile

Icon of a bar chart with an upward arrow.

Question 19

Given that $x^{2018}y^{2017} = \frac{1}{2}$, and $x^{2016}y^{2019} = 8$, then value of $x^2 + y^3$ is

- A $\frac{31}{4}$
B $\frac{35}{4}$
C $\frac{37}{4}$
D $\frac{33}{4}$

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Given that $x^{2018}y^{2017} = \frac{1}{2}$, and $x^{2016}y^{2019} = 8$, then value of $x^2 + y^3$ is

Handwritten solution: $\frac{(2)}{(1)} \Rightarrow \frac{y^2}{x^2} = 16$
 $y^2 = 16x^2$
 $y = 4x$
Substituting in (1): $x^{2018}(4x)^{2017} = \frac{1}{2}$
 $x^{2018} \cdot 4^{2017} \cdot x^{2017} = \frac{1}{2}$
 $x^{4035} \cdot 4^{2017} = \frac{1}{2}$
 $x^{4035} = \frac{1}{2 \cdot 4^{2017}} = \frac{1}{2^{4035}}$
 $x = \frac{1}{2}$
Substituting in (2): $\left(\frac{1}{2}\right)^{2016} (4x)^{2019} = 8$
 $\left(\frac{1}{2}\right)^{2016} \cdot 4^{2019} \cdot x^{2019} = 8$
 $\left(\frac{1}{2}\right)^{2016} \cdot 2^{4038} \cdot x^{2019} = 2^3$
 $2^{2022} \cdot x^{2019} = 2^3$
 $x^{2019} = \frac{1}{2^{2019}}$
 $x = \frac{1}{2}$
Now, $y = 4x = 2$
Value of $x^2 + y^3 = \left(\frac{1}{2}\right)^2 + 2^3 = \frac{1}{4} + 8 = \frac{33}{4}$

Video player interface with a play button and a person in the bottom right corner.

VIDEO SOLUTION

Question 20

The sum of all possible real values of x for which $\log_{x-3}(x^2 - 9) = \log_{x-3}(x + 1) + 2$, is

- A -3

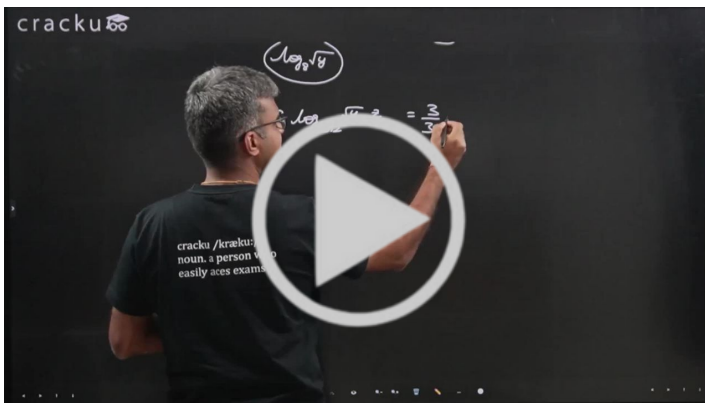
- B $\sqrt{33}$
- C 3
- D $\frac{3 + \sqrt{33}}{2}$

[VIEW SOLUTION](#)

Question 21

If $\log_{64} x^2 + \log_8 \sqrt{y} + 3 \log_{512} (\sqrt{y}z) = 4$, where x, y and z are positive real numbers, then the minimum possible value of $(x + y + z)$ is

- A 48
- B 36
- C 24
- D 96



[VIDEO SOLUTION](#)

CAT Percentile Predictor



Question 22

Suppose, $\log_3 x = \log_{12} y = a$, where x, y are positive numbers. If G is the geometric mean of x and y , and $\log_6 G$ is equal to

- A $\sqrt{a}$
- B $2a$
- C $a/2$
- D a

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Suppose, $\log_3 x = \log_{12} y = a$, where x, y are positive numbers. If G is the geometric mean of x and y , and $\log_6 G$ is equal to

$x = 3^a$
 $y = 12^a$
 $G = \sqrt{xy}$
 $\log_6 G$




VIDEO SOLUTION

Question 23

If $9^{2x-1} - 81^{x-1} = 1944$, then x is

- A 3
- B $9/4$
- C $4/9$
- D $1/3$

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If $9^{2x-1} - 81^{x-1} = 1944$, then x is

$9^{2x} = a$
 $9^{2x-1} = \frac{9^{2x}}{9} = \frac{a}{9}$
 $81^{x-1} = 9^{2x-2} = \frac{9^{2x}}{9^2} = \frac{a}{81}$
 $81^{x-1} = 9^{2x-2}$
 $9^{2x-1} - 9^{2x-2} = 1944$
 $9^{2x-2}(9 - 1) = 1944$
 $9^{2x-2} = \frac{1944}{8} = 243$
 $9^{2x-2} = 3^6$
 $2x-2 = 6$
 $2x = 8$
 $x = 4$




VIDEO SOLUTION

Question 24

Let x and y be positive real numbers such that

$\log_5 (x + y) + \log_5 (x - y) = 3$, and $\log_2 y - \log_2 x = 1 - \log_2 3$. Then xy equals

- A 150
- B 25
- C 100
- D 250

 VIDEO SOLUTION

Question 25

E 12^{1/12}

 VIDEO SOLUTION

B 5

C 4

D None of these

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If $\log_2 \log_7 (x^2 - x + 37) = 1$, then what could be the value of 'x'?

A) 3
C) 4
B) 5
D) None of these

The handwritten solution shows the equation $\log_2 \log_7 (x^2 - x + 37) = 1$. It simplifies to $\log_7 (x^2 - x + 37) = 2$, then $x^2 - x + 37 = 49$, leading to $x^2 - x - 12 = 0$. The roots are $x = 4$ and $x = -3$. The correct answer is C) 4.

VIDEO SOLUTION

Question 27

If $(x + 6\sqrt{2}) \frac{1}{2} - (x - 6\sqrt{2}) \frac{1}{2} = 2\sqrt{2}$, then x equals

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If $(x + 6\sqrt{2}) \frac{1}{2} - (x - 6\sqrt{2}) \frac{1}{2} = 2\sqrt{2}$, then x equals

Answer: _____

The handwritten solution shows the equation $(x + 6\sqrt{2}) \frac{1}{2} - (x - 6\sqrt{2}) \frac{1}{2} = 2\sqrt{2}$. It simplifies to $x + 6\sqrt{2} - x + 6\sqrt{2} = 4\sqrt{2}$, leading to $12\sqrt{2} = 4\sqrt{2}$, which is a contradiction. The correct answer is D) None of these.

VIDEO SOLUTION

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Question 28

If a, b and c are positive real numbers such that $a > 10 \geq b \geq c$ and $\frac{\log_8(a+b)}{\log_2 c} + \frac{\log_{27}(a-b)}{\log_3 c} = \frac{2}{3}$, then the greatest possible integer value of a is

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If a, b and c are positive real numbers such that $a > 10 \geq b \geq c$ and $\frac{\log_8(a+b)}{\log_2 c} + \frac{\log_2(a-b)}{\log_3 c} = \frac{2}{3}$, then the greatest possible integer value of a is

Answer: _____

The handwritten solution shows the following steps:

$$\log_8(a+b) = \frac{1}{3} \log_2(a+b)$$

$$\log_2(a-b) = \frac{1}{2} \log_2(a-b)$$

$$\log_2 c = \frac{1}{2} \log_2 c$$

$$\log_3 c = \frac{1}{3} \log_3 c$$

$$\frac{1}{3} \log_2(a+b) + \frac{1}{6} \log_2(a-b) = \frac{2}{3}$$

$$\log_2(a+b) + \frac{1}{2} \log_2(a-b) = 4$$

$$\log_2(a+b) + \log_2 \sqrt{a-b} = 4$$

$$\log_2(a+b) + \log_2 \sqrt{a-b} = \log_2 16$$

$$(a+b) \sqrt{a-b} = 16$$

$$a+b = \frac{16}{\sqrt{a-b}}$$

$$a = \frac{16}{\sqrt{a-b}} - b$$

$$a = \frac{16}{\sqrt{a-b}} - c$$

$$a = \frac{16}{\sqrt{a-b}} - c$$

$$a = \frac{16}{\sqrt{a-b}} - c$$

VIDEO SOLUTION

Question 29

If x is a positive real number such that $4 \log_{10} x + 4 \log_{100} x + 8 \log_{1000} x = 13$, then the greatest integer not exceeding x , is

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If x is a positive real number such that $4 \log_{10} x + 4 \log_{100} x + 8 \log_{1000} x = 13$, then the greatest integer not exceeding x , is

Answer: _____

The handwritten solution shows the following steps:

$$4 \log_{10} x + \frac{4}{2} \log_{10} x + \frac{8}{3} \log_{10} x = 13$$

$$4 \log_{10} x + 2 \log_{10} x + \frac{8}{3} \log_{10} x = 13$$

$$\left(4 + 2 + \frac{8}{3}\right) \log_{10} x = 13$$

$$\frac{20}{3} \log_{10} x = 13$$

$$\log_{10} x = \frac{13 \times 3}{20} = \frac{39}{20}$$

$$x = 10^{\frac{39}{20}}$$

$$x = 10^{1.95}$$

$$x = 10^{1.95}$$

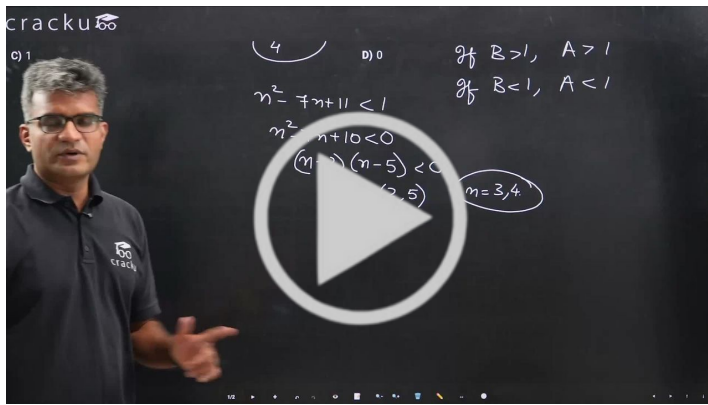
$$x = 10^{1.95}$$

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Question 30

The number of distinct integers n for which $\log_{\frac{1}{4}}(n^2 - 7n + 11) > 0$, is

- A 2
- B infinite
- C 1
- D 0

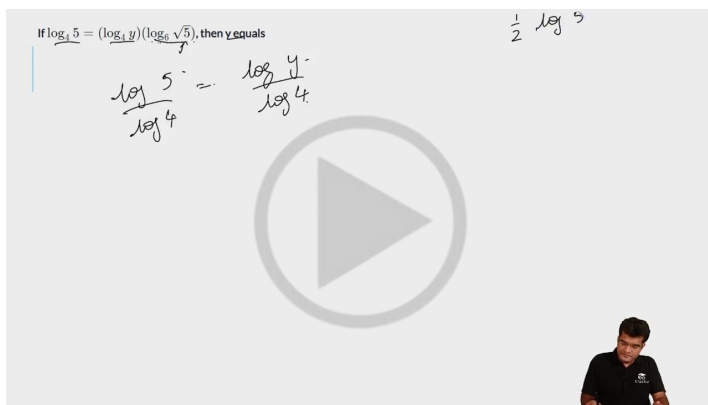


VIDEO SOLUTION



Question 31

If $\log_4 5 = (\log_4 y)(\log_6 \sqrt{5})$, then y equals



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Question 32

If x is a real number such that $\log_3 5 = \log_5 (2 + x)$, then which of the following is true?

- A $0 < x < 3$
- B $23 < x < 30$
- C $x > 30$
- D $3 < x < 23$

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If x is a real number such that $\log_3 5 = \log_5 (2 + x)$, then which of the following is true?

A) $0 < x < 3$
 B) $23 < x < 30$
 C) $x > 30$
 D) $3 < x < 23$

$\log_3 5 > \log_3 3 = 1$
 $\log_3 5 - \log_3 3 = 2$

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Question 33

If $\log_y x = (a * \log_z y) = (b * \log_x z) = ab$, then which of the following pairs of values for (a, b) is not possible?

- A $(-2, 1/2)$
- B $(1, 1)$
- C $(0.4, 2.5)$
- D $(\pi, 1/\pi)$
- E $(2, 2)$

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If $\log_y x = (a * \log_z y) = (b * \log_x z) = ab$, then which of the following pairs of values for (a, b) is not possible?

A) $(-2, 1/2)$
 B) $(1, 1)$
 C) $(0.4, 2.5)$
 D) $(\pi, 1/\pi)$
 E) $(2, 2)$

$\log_y x = a * \log_z y = b * \log_x z = ab$
 $\log_z y = b$
 $y^b = z^b$
 $y = z$
 $\log_y x = a$
 $x = y^a$
 $\log_x z = b$
 $z = x^b$
 $x = y^a = (x^b)^a = x^{ab}$
 $1 = x^{ab-a}$
 $1 = x^{a(b-1)}$
 $1 = x^{a(b-1)}$
 $1 = x^{a(b-1)}$

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CAT Daily Targets

Step by step, heading towards 99+ percentile

Question 34

If $x = -0.5$, then which of the following has the smallest value?

- A $2^{1/x}$
- B $1/x$

- C $1/x^2$
- D 2^x
- E $1/\sqrt{-x}$

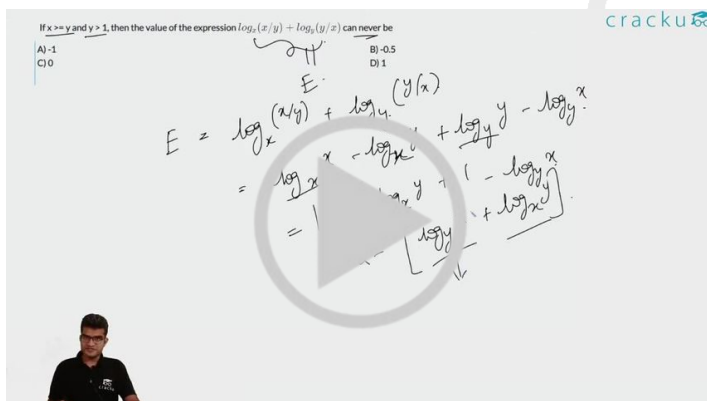


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Question 35

If $x \geq y$ and $y > 1$, then the value of the expression $\log_x(x/y) + \log_y(y/x)$ can never be

- A -1
- B -0.5
- C 0
- D 1



[VIDEO SOLUTION](#)

Question 36

The value of $\log_a(a/b) + \log_b(b/a)$, for $1 < a \leq b$ cannot be equal to

- A 0
- B -1
- C 1

D -0.5

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The value of $\log_a\left(\frac{b}{a}\right) + \log_b\left(\frac{a}{b}\right)$, for $1 < a \leq b$ cannot be equal to

A) 0
B) -1
C) 1
D) -0.5

Handwritten notes:
 $\log_a \frac{b}{a} = \log_a b - \log_a a = \log_a b - 1$
 $\log_b \frac{a}{b} = \log_b a - \log_b b = \log_b a - 1$
 $\log_a b = \frac{1}{\log_b a}$
 $\log_b a = \frac{1}{\log_a b}$

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Question 37

The real root of the equation $2^{6x} + 2^{3x+2} - 21 = 0$ is

- A $\log_2 9$
- B $\frac{\log_2 3}{3}$
- C $\log_2 27$
- D $\frac{\log_2 7}{3}$

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The real root of the equation $2^{6x} + 2^{3x+2} - 21 = 0$ is

A) $\log_2 9$
B) $\frac{\log_2 3}{3}$
C) $\log_2 27$
D) $\frac{\log_2 7}{3}$

Handwritten notes:
 $2^{6x} = a^2$
 $2^{3x+2} = 2^2 \times 2^{3x} = 4a$
 $a^2 + 4a - 21 = 0$
 $(a+7)(a-3) = 0$
 $a = 3$
 $2^{6x} = 3^2$
 $2^{3x} = 3$
 $3x = \log_2 3$
 $x = \frac{\log_2 3}{3}$

Video player interface with a play button and a person at a desk.

VIDEO SOLUTION

Question 38

If a, b, c are non-zero and $14^a = 36^b = 84^c$, then $6b\left(\frac{1}{c} - \frac{1}{a}\right)$ is equal to

 VIDEO SOLUTION

 VIDEO SOLUTION

D $-\log_2\left(\frac{1}{3}\right)$

If Y is a negative number such that $2^{Y^2(\log_2 5)} = 5^{\log_2 3}$, then Y equals to:

A) $\log_2\left(\frac{1}{3}\right)$ B) $\log_2\left(\frac{1}{5}\right)$
 C) $-\log_2\left(\frac{1}{3}\right)$ D) $-\log_2\left(\frac{1}{5}\right)$

Handwritten solution: $Y^2 = \frac{\log_2 3}{\log_2 5} \times \log_2 5$

Handwritten solution: $Y^2 = \frac{B}{A} \times \log_2 5$

VIDEO SOLUTION

Question 41

Find the value of $\frac{1}{1 + \frac{1}{3 - \frac{4}{2 + \frac{1}{3 - \frac{1}{2}}}}} + \frac{3}{3 - \frac{4}{3 + \frac{1}{2 - \frac{1}{2}}}}$

- A $\frac{13}{7}$
 B $\frac{15}{7}$
 C $\frac{11}{21}$
 D $\frac{17}{28}$

Find the value of $\frac{1}{1 + \frac{1}{3 - \frac{4}{2 + \frac{1}{3 - \frac{1}{2}}}}} + \frac{3}{3 - \frac{4}{3 + \frac{1}{2 - \frac{1}{2}}}}$

Handwritten solution: $\frac{4}{2 + \frac{1}{3}} = \frac{4}{\frac{7}{3}} = \frac{12}{7}$

Handwritten solution: $3 - \frac{12}{7} = \frac{21 - 12}{7} = \frac{9}{7}$

Handwritten solution: $\frac{3}{\frac{9}{7}} = \frac{21}{9} = \frac{7}{3}$

Handwritten solution: $\frac{1}{1 + \frac{1}{\frac{7}{3}}} = \frac{1}{1 + \frac{3}{7}} = \frac{1}{\frac{10}{7}} = \frac{7}{10}$

Handwritten solution: $\frac{7}{10} + \frac{7}{3} = \frac{7}{10} + \frac{24.5}{10} = \frac{31.5}{10} = \frac{63}{20}$

VIDEO SOLUTION

Question 42

$2^{73} - 2^{72} - 2^{71}$ is the same as

- A 2^{69}
 B 2^{70}

C 2^{71}

D 2^{72}



VIDEO SOLUTION



Question 43

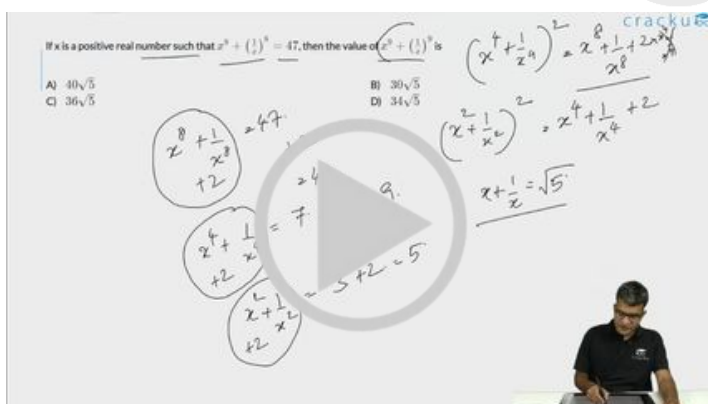
If x is a positive real number such that $x^8 + \left(\frac{1}{x}\right)^8 = 47$, then the value of $x^9 + \left(\frac{1}{x}\right)^9$ is

A $40\sqrt{5}$

B $30\sqrt{5}$

C $36\sqrt{5}$

D $34\sqrt{5}$



VIDEO SOLUTION

Question 44

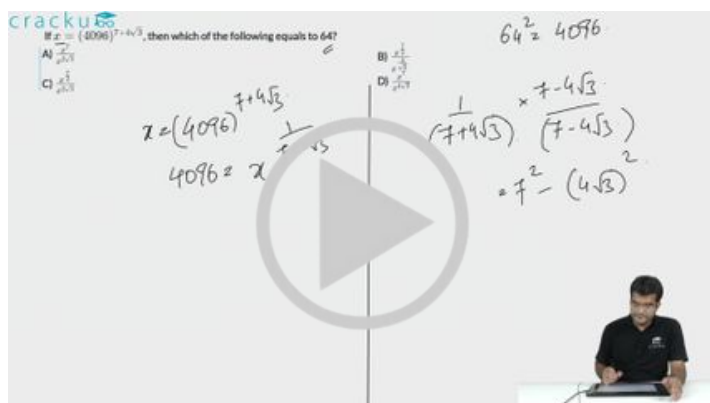
If $x = (4096)^{7+4\sqrt{3}}$, then which of the following equals to 64?

A $\frac{x^7}{x^{2\sqrt{3}}}$

B $\frac{x^{\frac{7}{2}}}{x^{\frac{4}{\sqrt{3}}}}$

C $\frac{x^{\frac{7}{2}}}{x^{2\sqrt{3}}}$

D $\frac{x^7}{x^{4\sqrt{3}}}$



VIDEO SOLUTION

Question 45

What are the values of x and y that satisfy both the equations?

$$2^{0.7x} * 3^{-1.25y} = 8\sqrt{6}/27$$

$$4^{0.3x} * 9^{0.2y} = 8 * 81^{1/5}$$

A $x = 2, y = 5$

B $x = 2.5, y = 6$

C $x = 3, y = 5$

D $x = 3, y = 4$

E $x = 5, y = 2$

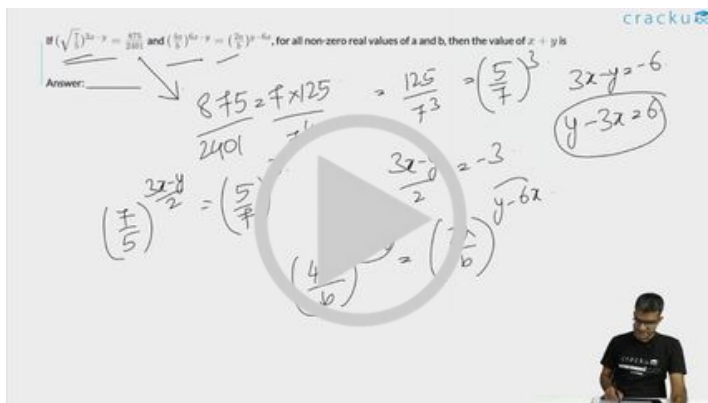
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100+ CAT Quant Questions



Question 46

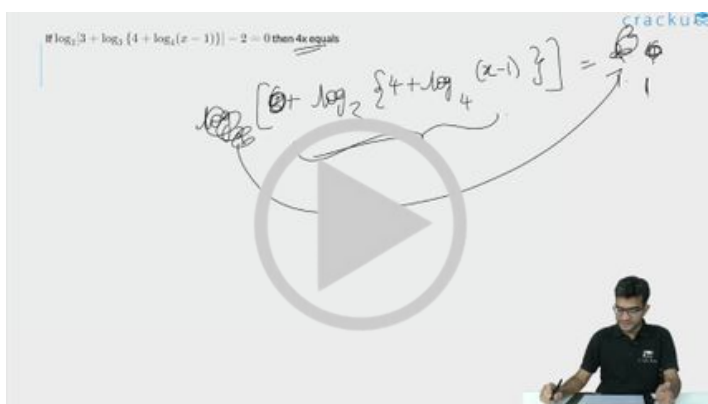
If $(\sqrt{\frac{7}{5}})^{3x-y} = \frac{875}{2401}$ and $(\frac{4a}{b})^{6x-y} = (\frac{2a}{b})^{y-6x}$, for all non-zero real values of a and b, then the value of $x + y$ is



VIDEO SOLUTION

Question 47

If $\log_2[3 + \log_3\{4 + \log_4(x - 1)\}] - 2 = 0$ then $4x$ equals



VIDEO SOLUTION

Question 48

Which of the following is true?

- A $7^{(3^2)} = (7^3)^2$
- B $7^{(3^2)} > (7^3)^2$
- C $7^{(3^2)} < (7^3)^2$
- D None of these

VIEW SOLUTION

100+ CAT DILR Questions

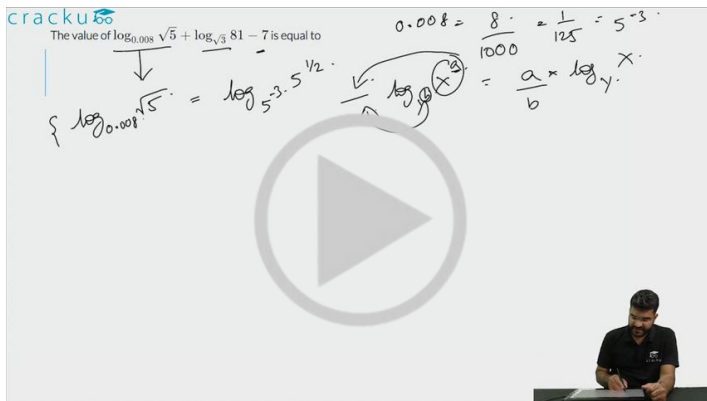
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Question 49

The value of $\log_{0.008} \sqrt{5} + \log_{\sqrt{3}} 81 - 7$ is equal to

- A $1/3$
- B $2/3$
- C $5/6$
- D $7/6$

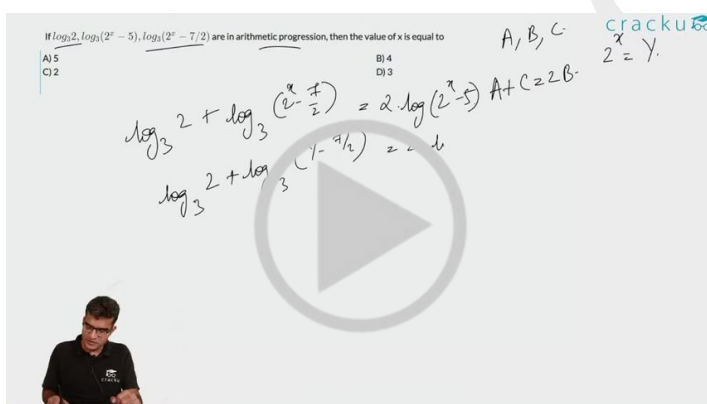


VIDEO SOLUTION

Question 50

If $\log_3 2, \log_3(2^x - 5), \log_3(2^x - 7/2)$ are in arithmetic progression, then the value of x is equal to

- A 5
- B 4
- C 2
- D 3



VIDEO SOLUTION

Question 51

For a real number a , if $\frac{\log_{15} a + \log_{32} a}{(\log_{15} a)(\log_{32} a)} = 4$ then a must lie in the range

- A $2 < a < 3$
- B $3 < a < 4$

C $4 < a < 5$

D $a > 5$

For a real number a , if $\frac{\log_{15} a + \log_{32} a}{(\log_{15} a)(\log_{32} a)} = 4$ then a must lie in the range

A) $2 < a < 3$ B) $3 < a < 4$
C) $4 < a < 5$ D) $a > 5$

Handwritten solution:
 $\frac{\log a}{\log 15} + \frac{\log a}{\log 32} = 4 \frac{\log a}{\log 15} \times \frac{\log a}{\log 32}$
 $\frac{\log a}{\log 15} + \frac{\log a}{\log 32} = 4 \frac{\log a}{\log 15} \times \frac{\log a}{\log 32}$



VIDEO SOLUTION

100+ CAT VARC Questions



Question 52

If $(a + b\sqrt{n})$ is the positive square root of $(29 - 12\sqrt{5})$, where a and b are integers, and n is a natural number, then the maximum possible value of $(a + b + n)$ is

A 18

B 22

C 4

D 6

If $(a + b\sqrt{n})$ is the positive square root of $(29 - 12\sqrt{5})$, where a and b are integers, and n is a natural number, then the maximum possible value of $(a + b + n)$ is

A) 18 B) 22
C) 4 D) 6

Handwritten solution:
 $(a + b\sqrt{n})^2 = 29 - 12\sqrt{5}$
 $a^2 + b^2n + 2ab\sqrt{n} = 29 - 12\sqrt{5}$
 $a^2 + b^2n = 29$
 $2ab\sqrt{n} = -12\sqrt{5}$
 $ab\sqrt{n} = -6\sqrt{5}$
 $a^2 + b^2n = 29$
 $a^2 + 180 = 29$
 $a^2 = -151$
 $b^2n = 180$
 $\frac{b^2n}{a^2} = \frac{180}{-151}$



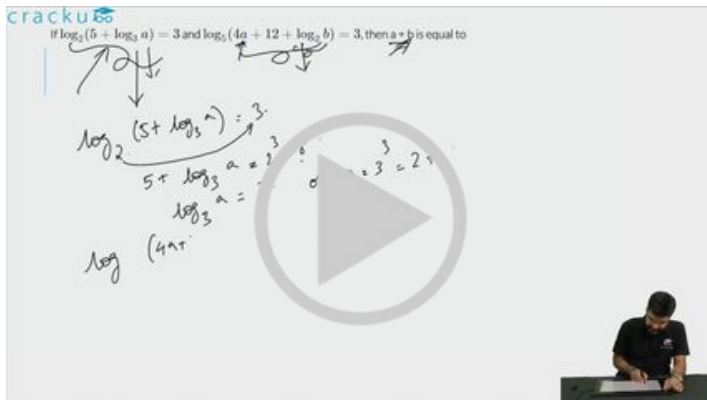
VIDEO SOLUTION

Question 53

If $\log_2(5 + \log_3 a) = 3$ and $\log_5(4a + 12 + \log_2 b) = 3$, then $a + b$ is equal to

A 59

- B 40
- C 32
- D 67



VIDEO SOLUTION

Question 54

If $\log_a 30 = A$, $\log_a\left(\frac{5}{3}\right) = -B$ and $\log_2 a = \frac{1}{3}$, then $\log_3 a$ equals

- A $\frac{2}{A+B-3}$
- B $\frac{2}{A+B} - 3$
- C $\frac{A+B}{2} - 3$
- D $\frac{A+B-3}{2}$



VIDEO SOLUTION

CAT Expected Questions PDF



Answers

1.C	2.B	3.B	4.10	5.3	6.C	7.C	8.7
9.24	10.A	11.2	12.99	13.A	14.B	15.A	16.D
17.A	18.D	19.D	20.D	21.A	22.D	23.B	24.A
25.B	26.C	27.11	28.14	29.31	30.D	31.36	32.D
33.E	34.B	35.D	36.C	37.B	38.3	39.A	40.B
41.B	42.C	43.D	44.C	45.E	46.14	47.5	48.B
49.C	50.D	51.C	52.A	53.A	54.A		

Explanations

1. **C**

Given, $\log_x(x^2 + 12) = 4$

$$\Rightarrow x^2 + 12 = x^4$$

$$\Rightarrow x^4 - x^2 - 12 = 0$$

$$\Rightarrow x^4 - 4x^2 + 3x^2 - 12 = 0$$

$$\Rightarrow x^2(x^2 - 4) + 3(x^2 - 4) = 0$$

$$\Rightarrow (x^2 - 4)(x^2 + 3) = 0 \Rightarrow \text{since, } x \text{ is a positive real number (given)} \Rightarrow x = 2.$$

Now, Given $3 \log_y x = 1$

$$\Rightarrow \log_y x = \frac{1}{3}$$

$$\Rightarrow x = y^{\frac{1}{3}}$$

$$\Rightarrow y = x^3 \Rightarrow y = 8.$$

$$\Rightarrow x + y = 2 + 8 = 10.$$

 VIDEO SOLUTION

2. **B**

Opening the square on the left-hand side, we get $a^2 + 3b^2 + 2ab\sqrt{3}$

Comparing the rational part on both sides we get: $a^2 + 3b^2 = 52$

And comparing the irrational part we get: $2ab\sqrt{3} = 30\sqrt{3}$

$ab = 15$, Since we are given that a and b are natural numbers, the possible values of a and b are $(1, 15)$, $(3, 5)$, $(5, 3)$, or $(15, 1)$

Putting these values in the first relation we got, we see that 15 squared would exceed the required value and would not be the case.

We need not check if $a=5, b=3$ or $a=3, b=5$ since the answer would be the same.

($a=5$ and $b=3$ would satisfy it)

$$a+b = 5+3 = 8$$

Therefore, Option B is the correct answer.

 VIDEO SOLUTION

3. **B**

$$x^u = 256$$

Taking log to the base 2 on both the sides,

$$u * \log_2 x = \log_2 256$$

$$\Rightarrow [(\log_2 x)^2 - 6\log_2 x + 12] * \log_2 x = 8$$

$$(\log_2 x)^3 - 6(\log_2 x)^2 + 12\log_2 x = 8$$

$$\text{Let } \log_2 x = t$$

$$t^3 - 6t^2 + 12t - 8 = 0$$

$$(t - 2)^3 = 0$$

$$\text{Therefore, } \log_2 x = 2$$

$$\Rightarrow x = 4 \text{ is the only solution}$$

Hence, option B is the correct answer.

 VIDEO SOLUTION

4. 10

It is given that $N^N = 2^{160}$

We can rewrite the equation as $N^N = (2^5)^{160/5} = 32^{32}$

$$\Rightarrow N = 32$$

$$N^2 + 2^N = 32^2 + 2^{32} = 2^{10} + 2^{32} = 2^{10} * (1 + 2^{22})$$

Hence, we can say that $N^2 + 2^N$ can be divided by 2^{10}

$$\text{Therefore, } x_{max} = 10$$

 VIDEO SOLUTION

5. 3

$$\log(2^a \times 3^b \times 5^c) = \frac{\log(2^2 \times 3^3 \times 5) + \log(2^6 \times 3 \times 5^7) + \log(2 \times 3^2 \times 5^4)}{3}$$

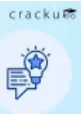
$$\log(2^a \times 3^b \times 5^c) = \frac{\log(2^{2+6+1} \times 3^{3+1+2} \times 5^{1+7+4})}{3}$$

$$\log(2^a \times 3^b \times 5^c) = \frac{\log(2^9 \times 3^6 \times 5^{12})}{3}$$

$$3\log(2^a \times 3^b \times 5^c) = \log(2^9 \times 3^6 \times 5^{12})$$

$$\text{Hence, } 3a = 9 \text{ or } a = 3$$

 VIDEO SOLUTION

 **CAT Important Topics** 

6. C

$$\text{We have, } (\sqrt{2})^{19} 3^4 4^{29} 8^n = 3^n 16^m (\sqrt[4]{64})$$

Converting both sides in powers of 2 and 3, we get

$$2^{\frac{19}{2}} 3^4 2^4 3^{2m} 2^{3n} = 3^n 2^{4m} 2^{\frac{6}{4}}$$

$$\text{Comparing the power of 2 we get, } \frac{19}{2} + 4 + 3n = 4m + \frac{6}{4}$$

$$\Rightarrow 4m = 3n + 12 \dots (1)$$

$$\text{Comparing the power of 3 we get, } 4 + 2m = n$$

Substituting the value of n in (1), we get

$$4m = 3(4+2m) + 12$$

$$\Rightarrow m = -12$$

 VIDEO SOLUTION

7. C

$$\text{Given, } \sqrt{5x+9} + \sqrt{5x-9} = 3(2 + \sqrt{2})$$

$$\Rightarrow \sqrt{5x+9} + \sqrt{5x-9} = 6 + 3\sqrt{2}$$

$$\Rightarrow \sqrt{5x+9} + \sqrt{5x-9} = \sqrt{36} + \sqrt{18}$$

Comparing the L.H.S. and R.H.S.

$$\Rightarrow 5x + 9 = 36 \Rightarrow 5x = 27 \Rightarrow x = \frac{27}{5} \text{ (can be verified using the second term as well).}$$

$$\Rightarrow \sqrt{10x+9} = \sqrt{\left(10 \times \frac{27}{5}\right) + 9} = \sqrt{63} = 3\sqrt{7}$$

 VIDEO SOLUTION

8. 7

It is given that $\log_{\sqrt{3}}(x) + \frac{\log_x(25)}{\log_x(0.008)} = \frac{16}{3}$, which can be written as:

$$\Rightarrow 2 \log_3 x + \log_{0.008} 25 = \frac{16}{3}$$

$$\Rightarrow 2 \log_3 x + \log_{\frac{8}{1000}} 25 = \frac{16}{3}$$

$$\Rightarrow 2 \log_3 x + \log_{\frac{1}{125}} 25 = \frac{16}{3}$$

$$\Rightarrow 2 \log_3 x + \log_{5^{-3}} (5)^2 = \frac{16}{3}$$

$$\Rightarrow 2 \log_3 x - \frac{2}{3} = \frac{16}{3}$$

$$\Rightarrow 2 \log_3 x = \frac{16}{3} + \frac{2}{3}$$

$$\Rightarrow 2 \log_3 x = 6$$

$$\Rightarrow \log_3 x^2 = 6 \Rightarrow x^2 = 3^6$$

$$\text{Hence, } \log_3 (3 \cdot x^2) = \log_3 (3 \cdot 3^6) = \log_3 3^7 = 7$$

 VIDEO SOLUTION

9. 24

$$\begin{aligned} & \frac{2 \times 4 \times 8 \times 16}{(\log_2 4)^2 (\log_4 8)^3 (\log_8 16)^4} \\ &= \frac{2^1 \times 2^2 \times 2^3 \times 2^4}{(\log_2 2^2)^2 (\log_2 2^3)^3 (\log_2 2^4)^4} \\ &= \frac{2^{1+2+3+4}}{(2)^2 \left(\frac{3}{2}\right)^3 \left(\frac{4}{2}\right)^4} \\ &= \frac{2^{10}}{4 \left(\frac{3}{2}\right)^3 \left(\frac{4}{2}\right)^4} = 24 \end{aligned}$$

 VIDEO SOLUTION

10. A

Taking $10^x = a$

we get $a + \frac{4}{a} = \frac{81}{2}$

This would give the quadratic equation: $2a^2 - 81a + 8 = 0$

We want to find the sum of possible values of x , let the value of x be x_1 and x_2

these would correspond to $\log a_1$, and $\log a_2$

The sum of $\log a_1 + \log a_2$ would be $\log (a_1 \times a_2)$

From the quadratic equation we got above, we can see that the product of the possible values of a would be $8/2 = 4$

Therefore, the sum of values of x would be $\log (4)$ which would be $2 \log_{10} 2$

Therefore, Option A is the correct answer.

 VIDEO SOLUTION



11.2

Taking a log for each of the expressions, we get the following:

$$\log_3 4 = a, \log_4 5 = b, \log_5 6 = c, \log_6 7 = d, \log_7 8 = e, \log_8 9 = f$$

The expression $abcfe$ would then be: $\log_3 4 \times \log_4 5 \times \log_5 6 \times \log_6 7 \times \log_7 8 \times \log_8 9$

Next, we can use this property of log: $\frac{\log_b a}{\log_b c} = \log_c a$

Using this, we get:

$$\frac{\log 4}{\log 3} \times \frac{\log 5}{\log 4} \times \frac{\log 6}{\log 5} \times \frac{\log 7}{\log 6} \times \frac{\log 8}{\log 7} \times \frac{\log 9}{\log 8}$$

All the terms will cancel out except: $\frac{\log 9}{\log 3} = \log_3 9 = 2$

Therefore, 2 is the correct answer.

 VIDEO SOLUTION

12.99

$$5 - \log_{10} \sqrt{1+x} + 4 \log_{10} \sqrt{1-x} = \log_{10} \frac{1}{\sqrt{1-x^2}}$$

We can re-write the equation as: $5 - \log_{10} \sqrt{1+x} + 4 \log_{10} \sqrt{1-x} = \log_{10} (\sqrt{1+x} \times \sqrt{1-x})^{-1}$

$$5 - \log_{10} \sqrt{1+x} + 4 \log_{10} \sqrt{1-x} = (-1) \log_{10} (\sqrt{1+x}) + (-1) \log_{10} (\sqrt{1-x})$$

$$5 = -\log_{10} \sqrt{1+x} + \log_{10} \sqrt{1+x} - \log_{10} \sqrt{1-x} - 4 \log_{10} \sqrt{1-x}$$

$$5 = -5 \log_{10} \sqrt{1-x}$$

$$\sqrt{1-x} = \frac{1}{10}$$

$$\text{Squaring both sides: } (\sqrt{1-x})^2 = \frac{1}{100}$$

$$\therefore x = 1 - \frac{1}{100} = \frac{99}{100}$$

$$\text{Hence, } 100x = 100 \times \frac{99}{100} = 99$$

 VIDEO SOLUTION

13.A

We have, $(5.55)^x = (0.555)^y = 1000$

Taking log in base 10 on both sides,

$$x(\log_{10} 555 - 2) = y(\log_{10} 555 - 3) = 3$$

$$\text{Then, } x(\log_{10} 555 - 2) = 3 \dots (1)$$

$$y(\log_{10} 555 - 3) = 3 \dots (2)$$

From (1) and (2)

$$\Rightarrow \log_{10} 555 = \frac{3}{x} + 2 = \frac{3}{y} + 3$$

$$\Rightarrow \frac{1}{x} - \frac{1}{y} = \frac{1}{3}$$

 VIDEO SOLUTION

14. B

If $f(x) = \log \frac{(1+x)}{(1-x)}$ then $f(y) = \log \frac{(1+y)}{(1-y)}$

Also $\text{Log}(A*B) = \text{Log} A + \text{Log} B$

$$f(x) + f(y) = \log \frac{(1+x)(1+y)}{(1-x)(1-y)}$$

$$= \log \frac{(1+xy+x+y)}{(1+xy-x-y)}$$

Dividing numerator and denominator by $(1+xy)$

$$\log \frac{\frac{(1+xy+x+y)}{1+xy}}{\frac{(1+xy-x-y)}{1+xy}}$$

$$= \log \frac{\frac{1+xy}{1+xy} + \frac{(x+y)}{1+xy}}{\frac{1+xy}{1+xy} - \frac{(x+y)}{1+xy}}$$

$$= \log \frac{1 + \frac{(x+y)}{(1+xy)}}{1 - \frac{(x+y)}{(1+xy)}}$$

Hence option B.

 VIDEO SOLUTION

15. A

Given that, $p^3 = q^4 = r^5 = s^6$

$$p^3 = s^6$$

$$p = s^{\frac{6}{3}} = s^2 \dots (1)$$

$$\text{Similarly, } q = s^{\frac{6}{4}} = s^{\frac{3}{2}} \dots (2)$$

$$\text{Similarly, } r = s^{\frac{6}{5}} \dots (3)$$

$$\Rightarrow \log_s(pqr)$$

By substituting value of p, q, and r from equation (1), (2) and (3)

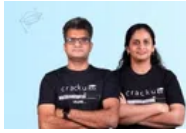
$$\Rightarrow \log_s(s^2 * s^{\frac{3}{2}} * s^{\frac{6}{5}})$$

$$\Rightarrow \log_s(s^{\frac{47}{10}})$$

$$\Rightarrow \frac{47}{10}$$

Hence, option A is the correct answer.

 VIDEO SOLUTION



CAT Mock Test

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16. D

Given that: $\log_{12} 81 = p$

$$\Rightarrow \log_{81} 12 = \frac{1}{p}$$

$$\Rightarrow \log_3 3 * 4 = \frac{4}{p}$$

$$\Rightarrow 1 + \log_3 4 = \frac{4}{p}$$

Using Componendo and Dividendo,

$$\Rightarrow \frac{1 + \log_3 4 - 1}{1 + \log_3 4 + 1} = \frac{4 - p}{4 + p}$$

$$\Rightarrow \frac{\log_3 4}{2 + \log_3 4} = \frac{4 - p}{4 + p}$$

$$\Rightarrow \frac{\log_3 4}{\log_3 9 + \log_3 4} = \frac{4 - p}{4 + p}$$

$$\Rightarrow \frac{\log_3 4}{\log_3 36} = \frac{4 - p}{4 + p}$$

$$\Rightarrow 3 * \frac{4 - p}{4 + p} = \frac{3 \log_3 4}{\log_3 36}$$

$$\Rightarrow 3 * \frac{4 - p}{4 + p} = \frac{\log_3 64}{\log_3 36}$$

$$\Rightarrow 3 * \frac{4 - p}{4 + p} = \log_{36} 64$$

$$\Rightarrow 3 * \frac{4 - p}{4 + p} = \log_{6^2} 8^2 = \log_6 8. \text{ Hence, option D is the correct answer.}$$

VIDEO SOLUTION

17. A

We know that $\frac{1}{\log_a b} = \frac{\log_x a}{\log_x b}$

Therefore, we can say that $\frac{1}{\log_2 100} = \frac{\log_{10} 2}{\log_{10} 100}$

$$\Rightarrow \frac{1}{\log_2 100} - \frac{1}{\log_4 100} + \frac{1}{\log_5 100} - \frac{1}{\log_{10} 100} + \frac{1}{\log_{20} 100} - \frac{1}{\log_{25} 100} + \frac{1}{\log_{50} 100}$$

$$\Rightarrow \frac{\log_{10} 2}{\log_{10} 100} - \frac{\log_{10} 4}{\log_{10} 100} + \frac{\log_{10} 5}{\log_{10} 100} - \frac{\log_{10} 10}{\log_{10} 100} + \frac{\log_{10} 20}{\log_{10} 100} - \frac{\log_{10} 25}{\log_{10} 100} + \frac{\log_{10} 50}{\log_{10} 100}$$

We know that $\log_{10} 100 = 2$

$$\Rightarrow \frac{1}{2} * [\log_{10} 2 - \log_{10} 4 + \log_{10} 5 - \log_{10} 10 + \log_{10} 20 - \log_{10} 25 + \log_{10} 50]$$

$$\Rightarrow \frac{1}{2} * [\log_{10} \frac{2 * 5 * 20 * 50}{4 * 10 * 25}]$$

$$\Rightarrow \frac{1}{2} * [\log_{10} 10]$$

$$\Rightarrow \frac{1}{2}$$

 VIDEO SOLUTION

18. D

Given that: $2^x = 3^{\log_5 2}$

$$\Rightarrow 2^x = 2^{\log_5 3}$$

$$\Rightarrow x = \log_5 3$$

$$\Rightarrow x = \log_5 \frac{3 \cdot 5}{5}$$

$$\Rightarrow x = \log_5 5 + \log_5 \frac{3}{5}$$

$$\Rightarrow x = 1 + \log_5 \frac{3}{5}. \text{ Hence, option D is the correct answer.}$$

 VIDEO SOLUTION

19. D

Given that $x^{2018}y^{2017} = \frac{1}{2} \dots (1)$

$$x^{2016}y^{2019} = 8 \dots (2)$$

Equation (2)/ Equation (1)

$$\frac{y^2}{x^2} = \frac{8}{1/2}$$

$$\frac{y}{x} = 4 \text{ or } -4$$

Case 1: When $\frac{y}{x} = 4$

$$x^{2018}(4x)^{2017} = \frac{1}{2}$$

$$x^{2018+2017}(2)^{4034} = \frac{1}{2}$$

$$x^{4035} = \frac{1}{(2)^{4035}}$$

$$x = \frac{1}{2}$$

Since, $\frac{y}{x} = 4, \Rightarrow y = 2$

$$\text{Therefore, } x^2 + y^3 = \frac{1}{4} + 8 = \frac{33}{4}$$

Case 2: When $\frac{y}{x} = -4$

$$x^{2018}(-4x)^{2017} = \frac{1}{2}$$

$$x^{2018+2017}(2)^{4034} = \frac{-1}{2}$$

$$x^{4035} = \frac{1}{(-2)^{4035}}$$

$$x = \frac{-1}{2}$$

Since, $\frac{y}{x} = -4, \Rightarrow y = 2$

Therefore, $x^2 + y^3 = \frac{1}{4} + 8 = \frac{33}{4}$. Hence, option D is the correct answer.

 VIDEO SOLUTION

20. D

Since the base of the logarithm has to be greater than zero and cannot be 1, x has to be greater than 3 and cannot be 4. Also, since $x^2 - 9 > 0$, x will be greater than 3.

The equation can be rewritten as;

$$\log_{x-3} (x^2 - 9) - \log_{x-3} (x + 1) = 2$$

Or,

$$\log_{x-3} \frac{x^2 - 9}{x + 1} = 2$$

$$\Rightarrow \frac{(x + 3)(x - 3)}{x + 1} = (x - 3)^2$$

$$\Rightarrow \frac{(x + 3)}{x + 1} = (x - 3)$$

$$\Rightarrow (x + 3) = (x - 3)(x + 1)$$

$$\Rightarrow x + 3 = x^2 - 3x + x - 3$$

$$\Rightarrow x^2 - 3x - 6 = 0$$

The roots of the quadratic above are $\frac{3 \pm \sqrt{3^2 + 24}}{2}$

The negative value of x will not be possible, the positive value that satisfies is $\frac{3 + \sqrt{33}}{2}$

The correct answer is option D.

 VIEW SOLUTION



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21. A

$$64 = 8^2 \text{ and } 512 = 8^3$$

$$\log_{64} x^2 + \log_8 \sqrt{y} + 3 \log_{512} (\sqrt{y}z) = 4,$$

Property of log: $\log_{b^m} a^n = \frac{n}{m} \log_b a$

$$\log_{8^2} x^2 + \log_8 \sqrt{y} + 3 \log_{8^3} (\sqrt{y}z) = 4$$

Using the above-mentioned property, the expression becomes $\log_8 x + \log_8 \sqrt{y} + \log_8 (\sqrt{y}z) = 4$

$$\log_8 x \sqrt{y} \cdot (\sqrt{y}z) = 4$$

$$\log_8 xyz = 4$$

$$xyz = 8^4 = 2^{12}$$

Using AM-GM inequality

$$\frac{(x+y+z)}{3} \geq \sqrt[3]{xyz}$$

$$\frac{(x+y+z)}{3} \geq 2^4$$

$$(x + y + z) \geq 48$$

 VIDEO SOLUTION

22. **D**

We know that $\log_3 x = a$ and $\log_{12} y = a$

Hence, $x = 3^a$ and $y = 12^a$

Therefore, the geometric mean of x and y equals $\sqrt{x \times y}$

This equals $\sqrt{3^a \times 12^a} = 6^a$

Hence, $G = 6^a$ Or, $\log_6 G = a$

 VIDEO SOLUTION

23. **B**

$$\frac{81^x}{9} - \frac{81^x}{81} = 1944$$

$$81^x * \left[\frac{1}{9} - \frac{1}{81}\right] = 1944$$

$$81^x * \left[\frac{1}{81}\right] = 243$$

$$3^{4x} = 3^9$$

$$x = \frac{9}{4}$$

 VIDEO SOLUTION

24. **A**

We have, $\log_5 (x + y) + \log_5 (x - y) = 3$

$$\Rightarrow x^2 - y^2 = 125 \dots (1)$$

$$\log_2 y - \log_2 x = 1 - \log_2 3$$

$$\Rightarrow \frac{y}{x} = \frac{2}{3}$$

$$\Rightarrow 2x = 3y \Rightarrow x = \frac{3y}{2}$$

On substituting the value of x in 1, we get

$$\frac{5x^2}{4} = 125$$

$$\Rightarrow y = 10, x = 15$$

Hence $xy = 150$

 VIDEO SOLUTION

25. **B**

Make the power equal and compare the denominators.

$2^{1/2}$ can be written as $64^{1/12}$

$3^{1/3}$ can be written as $81^{1/12}$

$4^{1/4}$ can be written as $64^{1/12}$

$6^{1/6}$ can be written as $36^{1/12}$

Among these, $81^{1/12}$ is the greatest $\Rightarrow 3^{1/3}$ is the greatest.



26. C

$$\log_2 \log_7 (x^2 - x + 37) = 1$$

$$\log_7 (x^2 - x + 37) = 2$$

$$(x^2 - x + 37) = 7^2$$

$$\text{Given eq. can be reduced to } x^2 - x + 37 = 49$$

So x can be either -3 or 4.

27. 11

Squaring on both sides, we get:

$$x + 6\sqrt{2} + x - 6\sqrt{2} - 2(x^2 - 72)^{\frac{1}{2}} = 8$$

$$x - (x^2 - 72)^{\frac{1}{2}} = 4$$

Bringing x to the other side, we get:

$$-(x^2 - 72)^{\frac{1}{2}} = 4 - x$$

Squaring on both sides again, we get:

$$x^2 - 72 = 16 + x^2 - 8x$$

$$8x = 88$$

$$x = 11$$

Therefore, 11 is the correct answer.

28. 14

The first term of the expression can be rewritten as $\frac{\frac{1}{3} \log_2 (a+b)}{\log_2 c}$

Using the property $\frac{m}{n} \log_a b = \log_a b^{\frac{m}{n}}$ this can be rewritten as

$$\frac{\log_2 (a+b)^{\frac{1}{3}}}{\log_2 c}$$

And finally using the property $\frac{\log_b a}{\log_b c} = \log_c a$, we can rewrite the expression as

$$\log_c (a+b)^{\frac{1}{3}}$$

Doing identical operations in the second term, we get the entire left-hand side to be:

$$\log_c (a+b)^{\frac{1}{3}} + \log_c (a-b)^{\frac{1}{3}}$$

Using property $\log_c a + \log_c b = \log_c (ab)$ we get

$$\log_c \left[(a+b)^{\frac{1}{3}} (a-b)^{\frac{1}{3}} \right]$$

$$\log_c \left[(a+b)(a-b) \right]^{\frac{1}{3}}$$

$$\log_c \left[(a^2 - b^2) \right]^{\frac{1}{3}}$$

This expression is given to be equal to $2/3$

Using the definition of log: $\log_c N = a$ which is $c^a = N$

we get: $c^{\frac{2}{3}} = (a^2 - b^2)^{\frac{1}{3}}$

Cubing both sides:

$$c^2 = a^2 - b^2$$

Finally giving $a^2 = b^2 + c^2$

We have upper limits on b and c as 10, and we want to maximize the value of a squared.

This can be thought of as a right-angled triangle, and the value of a will be maximum when both b and c are equal to 10, giving $a^2 = 200$, but this would not give an integer value of a

We need to adjust a^2 to the biggest square less than 200, which is 196

Giving the value of a as 14.

Therefore, 14 is the correct answer.

 VIDEO SOLUTION

29.31

Using the logarithmic property that $\log_{a^p} b = \frac{1}{p} \log_a b$

$$4 \log_{10} x + 4 \log_{100} x + 8 \log_{1000} x = 13$$

Can be written as

$$4 \log_{10} x + 2 \log_{10} x + \frac{8}{3} \log_{10} x = 13$$

$$\frac{26}{3} \log_{10} x = 13$$

$$\log_{10} x = 1.5$$

$$x = 10^{1.5}$$

$$x = \sqrt{1000}$$

$$[\sqrt{1000}] = 31$$

Where $[\cdot]$ is Greatest Integer Function since that is what is asked in the question, 31 is the greatest integer that does not exceed x.

 VIDEO SOLUTION

30.D

For base of log in range $1/4 \in (0, 1)$ and $\log_{1/4}(x) > 0$ is true only if $0 < x < 1$.

For integer n, $x = n^2 - 7n + 11$ is an integer, so it cannot lie strictly between 0 and 1.

So, there is no integer value for which this inequality is satisfied.

 VIDEO SOLUTION



CAT Syllabus PDF



31.36

$$\frac{\log 5}{2 \log 2} = \frac{\log y}{2 \log 2} \cdot \frac{\log 5}{2 \log 6}$$

$$\log 36 = \log y; \therefore y = 36$$

 VIDEO SOLUTION

32.D

$$1 < \log_3 5 < 2$$

$$\Rightarrow 1 < \log_5(2+x) < 2$$

$$\Rightarrow 5 < 2+x < 25$$

$$\Rightarrow 3 < x < 23$$

[VIDEO SOLUTION](#)

33. E

$$\log_y x = ab$$

$$a * \log_z y = ab \Rightarrow \log_z y = b$$

$$b * \log_x z = ab \Rightarrow \log_x z = a$$

$$\log_y x = \log_z y * \log_x z \Rightarrow \log x / \log y = \log y / \log z * \log z / \log x$$

$$\Rightarrow \frac{\log x}{\log y} = \frac{\log y}{\log x}$$

$$\Rightarrow (\log x)^2 = (\log y)^2$$

$$\Rightarrow \log x = \log y \text{ or } \log x = -\log y$$

So, $x = y$ or $x = 1/y$

So, $ab = 1$ or -1

Option 5) is not possible

[VIDEO SOLUTION](#)

34. B

2^p is always positive

x^2 is always non negative.

$1/\sqrt{-x}$ is always positive.

$\frac{1}{x}$ is negative when x is negative.

In this case, x is negative $\Rightarrow \frac{1}{x}$ is smallest.

[VIDEO SOLUTION](#)

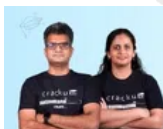
35. D

$$\log_x(x/y) + \log_y(y/x) = 1 - \log_x(y) + 1 - \log_y(x)$$

$$= 2 - (\log_x y + 1/\log_x y) \leq 0 \text{ (Since } \log_x y + 1/\log_x y \geq 2)$$

So, the value of the expression cannot be 1.

[VIDEO SOLUTION](#)



CAT Formula PDF



36. C

On expanding the expression we get $1 - \log_a b + 1 - \log_b a$

$$\text{or } 2 - \left(\log_a b + \frac{1}{\log_a b} \right)$$

Now applying the property of $AM \geq GM$, we get that $\frac{(\log_a b + \frac{1}{\log_a b})}{2} \geq 1$ or $\left(\log_a b + \frac{1}{\log_a b} \right) \geq 2$ Hence from here we can conclude that the expression will always be equal to 0 or less than 0. Hence any positive value is not possible. So 1 is not possible.

VIDEO SOLUTION

37. B

Let $2^{3x} = v$

$$2^{6x} + 2^{3x+2} - 21 = 0$$

$$= v^2 + 4v - 21 = 0$$

$$= (v+7)(v-3) = 0$$

$$v = 3, -7$$

$$2^{3x} = 3 \text{ or } 2^{3x} = -7 \text{ (This can be negated)}$$

$$3x = \log_2 3$$

$$x = \log_2 3/3$$

VIDEO SOLUTION

38. 3

Let $14^a = 36^b = 84^c = k$

$$\Rightarrow a = \log_{14} k, b = \log_{36} k, c = \log_{84} k$$

$$6b\left(\frac{1}{c} - \frac{1}{a}\right) = 6 \cdot \frac{1}{2} \log_6 k (\log_k 84 - \log_k 14) = 3$$

VIDEO SOLUTION

39. A

$$\text{It is given that } 9^{x-\frac{1}{2}} - 2^{2x-2} = 4^x - 3^{2x-3}$$

Let us try to reduce them to powers of 3 and 2

$$\text{The given equation can be reduced to } 3^{2x-1} + 3^{2x-3} = 2^{2x} + 2^{2x-2}$$

$$\text{Hence, } 3^{2x-3} \times 10 = 2^{2x-2} \times 5$$

$$\text{Therefore, } 3^{2x-3} = 2^{2x-3}$$

$$\text{This is possible only if } 2x - 3 = 0 \text{ or } x = 3/2$$

VIDEO SOLUTION

40. B

$$2^{Y^2(\log_3 5)} = 5^{Y^2(\log_3 2)}$$

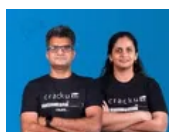
$$\text{Given, } 5^{Y^2(\log_3 2)} = 5^{(\log_2 3)}$$

$$\Rightarrow Y^2 (\log_3 2) = (\log_2 3) \Rightarrow Y^2 = (\log_2 3)^2$$

$$\Rightarrow Y = (-\log_2 3) \text{ or } (\log_2 3)$$

$$\text{since } Y \text{ is a negative number, } Y = (-\log_2 3) = (\log_2 \frac{1}{3})$$

VIDEO SOLUTION



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Accurate analysis of your profile



41. B

$$\begin{aligned}
& 1 + \frac{1}{3 - \frac{1}{2 + \frac{1}{3 - \frac{1}{2}}}} \\
&= \frac{1}{1 + \frac{1}{3 - \frac{1}{2 + \frac{1}{3 - \frac{1}{2}}}}} \\
&= \frac{1}{1 + \frac{1}{3 - \frac{4}{2 + \frac{2}{5}}}} \\
&= \frac{1}{1 + \frac{1}{3 - \frac{20}{12}}} \\
&= \frac{1}{1 + \frac{1}{3 - \frac{20}{12}}} \\
&= \frac{1}{1 + \frac{12}{16}} \\
&= \frac{1}{\frac{28}{16}} \\
&= \frac{16}{28}
\end{aligned}$$

Similarly, Second term can be reduced to $\frac{33}{21}$ or $\frac{11}{7}$

$$\text{Sum will be} = \frac{11}{7} + \frac{4}{7} = \frac{15}{7}$$

 VIDEO SOLUTION

42. C

$$\begin{aligned}
& 2^{71}(2^2 - 2^1 - 1) \\
& 2^{71}(4 - 2 - 1) \\
& 2^{71}
\end{aligned}$$

 VIDEO SOLUTION

43. D

It is given that $x^8 + \left(\frac{1}{x}\right)^8 = 47$, which can be written as:

$$\begin{aligned}
& \Rightarrow (x^4)^2 + \left(\frac{1}{x^4}\right)^2 = 47 \\
& \Rightarrow \left(x^4 + \frac{1}{x^4}\right)^2 - 2 \cdot x^4 \cdot \frac{1}{x^4} = 47 \\
& \Rightarrow \left(x^4 + \frac{1}{x^4}\right)^2 = 49 \\
& \Rightarrow x^4 + \frac{1}{x^4} = 7
\end{aligned}$$

Similarly, $x^4 + \frac{1}{x^4} = 7$ can be expressed as:

$$\begin{aligned}
& \Rightarrow (x^2)^2 + \left(\frac{1}{x^2}\right)^2 = 7 \\
& \Rightarrow \left(x^2 + \frac{1}{x^2}\right)^2 - 2 \cdot x^2 \cdot \frac{1}{x^2} = 7 \\
& \Rightarrow \left(x^2 + \frac{1}{x^2}\right)^2 = 9 \\
& \Rightarrow x^2 + \frac{1}{x^2} = 3
\end{aligned}$$

By the same logic, we get $x + \frac{1}{x} = \sqrt{5}$

$$\text{Now, } x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3 \cdot x \cdot \frac{1}{x} \left(x + \frac{1}{x}\right)$$

$$\Rightarrow x^3 + \frac{1}{x^3} = (\sqrt{5})^3 - 3\sqrt{5} = 2\sqrt{5}$$

By the same logic, we can say that

$$\Rightarrow x^9 + \frac{1}{x^9} = \left(x^3 + \frac{1}{x^3}\right)^3 - 3 \cdot x^3 \cdot \frac{1}{x^3} \left(x^3 + \frac{1}{x^3}\right)$$

$$\Rightarrow x^9 + \frac{1}{x^9} = (2\sqrt{5})^3 - 3(2\sqrt{5})$$

$$\Rightarrow x^9 + \frac{1}{x^9} = 40\sqrt{5} - 6\sqrt{5} = 34\sqrt{5}$$

The correct option is D

[VIDEO SOLUTION](#)

44. C

$$x = 2^{12(7+4\sqrt{3})}$$

$$x^{\frac{7}{2}} = 2^{42(7+4\sqrt{3})}$$

$$x^{2\sqrt{3}} = 2^{24\sqrt{3}(7+4\sqrt{3})}$$

$$\frac{x^{\frac{7}{2}}}{x^{2\sqrt{3}}} = 2^{(7+4\sqrt{3})(42-24\sqrt{3})} = 2^{(7+4\sqrt{3})(7-4\sqrt{3})6} = 2^6$$

Hence C is correct answer.

[VIDEO SOLUTION](#)

45. E

$$2^{0.7x} * 3^{-1.25y} = 8\sqrt{6}/27 \Rightarrow 2^{0.7x} * 3^{-1.25y} = 2^{3.5} * 3^{-2.5}$$

$$\Rightarrow 0.7x = 3.5 \Rightarrow x = 5$$

$$\Rightarrow -1.25y = -2.5 \Rightarrow y = 2$$

$$4^{0.3x} * 9^{0.2y} = 8 * 81^{1/5} \Rightarrow 2^{0.6x} * 3^{0.4y} = 2^3 * 3^{0.8}$$

$$\Rightarrow 0.6x = 3 \Rightarrow x = 5$$

$$\Rightarrow 0.4y = 0.8 \Rightarrow y = 2$$

$$\Rightarrow (5, 2) \text{ is the solution.}$$

[VIEW SOLUTION](#)

CAT Percentile Predictor



46. 14

$$\left(\sqrt{\frac{7}{5}}\right)^{3x-y} = \frac{875}{2401}$$

$$\left(\frac{7}{5}\right)^{\frac{(3x-y)}{2}} = \frac{125}{343}$$

$$\left(\frac{7}{5}\right)^{\frac{(3x-y)}{2}} = \left(\frac{7}{5}\right)^{-3}$$

$$3x-y = -6$$

$$\left(\frac{4a}{b}\right)^{6x-y} = \left(\frac{2a}{b}\right)^{y-6x}$$

Therefore, $y=6x$ as the bases are different so the power should be zero for the results to be equal.

$$3x-y=-6$$

$$\text{or, } 3x - 6x = -6$$

$$\text{or } x = 2$$

$$y = 6x = 12$$

$$x+y = 14$$

 VIDEO SOLUTION

47.5

We have :

$$\log_2 \{3 + \log_3 \{4 + \log_4 (x - 1)\}\} = 2$$

$$\text{we get } 3 + \log_3 \{4 + \log_4 (x - 1)\} = 4$$

$$\text{we get } \log_3 (4 + \log_4 (x - 1)) = 1$$

$$\text{we get } 4 + \log_4 (x - 1) = 3$$

$$\log_4 (x - 1) = -1$$

$$x-1 = 4^{-1}$$

$$x = \frac{1}{4} + 1 = \frac{5}{4}$$

$$4x = 5$$

 VIDEO SOLUTION

48. B

$$7^{(3^2)} = 7^9$$

$$(7^3)^2 = 7^6$$

$$\text{So } 7^{(3^2)} > (7^3)^2$$

 VIEW SOLUTION

49. C

$$\log_{0.008} \sqrt{5} + \log_{\sqrt{3}} 81 - 7$$

$$81 = 3^4 \text{ and } 0.008 = \frac{8}{1000} = \frac{2^3}{10^3} = \frac{1}{5^3} = 5^{-3}$$

Hence,

$$\log_{0.008} \sqrt{5} + 8 - 7$$

$$\log_{5^{-3}} 5^{\frac{1}{2}} + 8 - 7$$

$$\frac{\log 5^{0.5}}{\log 5^{-3}} + 1$$

$$-\frac{1}{6} + 1$$

$$= \frac{5}{6}$$

 VIDEO SOLUTION

50. D

$$2\log(2^x - 5) = \log 2 + \log(2^x - 7/2)$$

$$\text{Let } 2^x = t$$

$$\Rightarrow (t - 5)^2 = 2(t - 7/2)$$

$$\Rightarrow t^2 + 25 - 10t = 2t - 7$$

$$\Rightarrow t^2 - 12t + 32 = 0$$

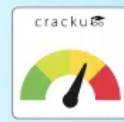
$$\Rightarrow t = 8, 4$$

Therefore, $x = 2$ or 3 , but $2^x > 5$, so $x = 3$

 VIDEO SOLUTION



CAT Score Calculator



51. C

$$\text{We have : } \frac{\log_{15} a + \log_{32} a}{(\log_{15} a)(\log_{32} a)} = 4$$

$$\text{We get } \frac{\left(\frac{\log a}{\log 15} + \frac{\log a}{\log 32}\right)}{\frac{\log a}{\log 15} \times \frac{\log a}{\log 32}} = 4$$

$$\text{we get } \log a (\log 32 + \log 15) = 4 (\log a)^2$$

$$\text{we get } (\log 32 + \log 15) = 4 \log a$$

$$= \log 480 = \log a^4$$

$$= a^4 = 480$$

so we can say a is between 4 and 5 .

[VIDEO SOLUTION](#)

52. A

$(a + b\sqrt{n})$ is the positive square root of $(29 - 12\sqrt{5})$

$$\text{So } 29 - 12\sqrt{5} = (a + b\sqrt{n})^2$$

$$29 - 12\sqrt{5} = a^2 + b^2n + 2ab\sqrt{n}$$

$$a^2 + b^2n = 29 \text{ and}$$

$$ab\sqrt{n} = -6\sqrt{5}$$

$$a^2b^2n = 180$$

$$b^2n = \frac{180}{a^2}$$

Substituting this in the above equation,

$$a^2 + \frac{180}{a^2} = 29$$

$$a^4 - 29a^2 + 180 = 0$$

$$a^2 = \frac{(29 \pm \sqrt{29^2 - 4(180)})}{2}$$

$$a^2 = \frac{(29 + \sqrt{841 - 720})}{2}$$

$$a^2 = 9 \text{ or } 20$$

That means, one of a^2 or b^2n is 9 and 20.

We also have, $ab\sqrt{n} = -6\sqrt{5}$ that means one of a or b should be negative

And also the fact that this is a positive square root,

And we need to maximise the value of a, b and n.

We can have $a=3$, $b=1$ and $n=20$.

This satisfies all the above equations, and the value of $a+b+n=18$.

[VIDEO SOLUTION](#)

53. A

$$\log_2(5 + \log_3 a) = 3$$

$$\Rightarrow 5 + \log_3 a = 8$$

$$\Rightarrow \log_3 a = 3$$

$$\text{or } a = 27$$

$$\log_5(4a + 12 + \log_2 b) = 3$$

$$\Rightarrow 4a + 12 + \log_2 b = 125$$

Putting $a = 27$, we get

$$\log_2 b = 5$$

$$\text{or, } b = 32$$

$$\text{So, } a + b = 27 + 32 = 59$$

Hence, option A is the correct answer.

 VIDEO SOLUTION

54. **A**

$$\log_a 30 = A \text{ or } \log_a 5 + \log_a 2 + \log_a 3 = A \dots\dots\dots(1)$$

$$\log_a \left(\frac{5}{3}\right) = -B \text{ or } \log_a 3 - \log_a 5 = B \dots\dots\dots(2)$$

$$\text{and finally } \log_a 2 = 3$$

$$\text{Substituting this in (1) we get } \log_a 5 + \log_a 3 = A - 3$$

Now we have two equations in two variables (1) and (2) . On solving we get

$$\log_a 3 = \frac{(A+B-3)}{2} \text{ or } \log_3 a = \frac{2}{A+B-3}$$

 VIDEO SOLUTION

